

ANGELINA COUNTY AP, TX, USA

WMO: 722446

Lat: 31.236N

Lon: 94.754W

Elev: 288

StdP: 14.54

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93987

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.0	29.9	13.7	11.2	36.5	18.2	13.9	40.5	19.8	57.4	18.3	58.4	3.5	0	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.1	99.0	76.0	96.4	76.2	94.3	76.3	79.8	90.1	79.1	89.2	78.4	88.3	6.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	143.7	82.6	76.7	140.7	82.3	75.9	136.7	81.8	43.3	89.5	42.6	89.3	41.9	88.7	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.6	15.9	13.8	DB	21.0	101.3	3.4	3.3	18.6	103.8	16.6	105.7	14.7	107.6	12.3	110.0	
			WB	19.3	81.0	3.0	1.0	17.2	81.7	15.4	82.3	13.7	82.9	11.5	83.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.6	49.9	53.9	60.4	67.3	75.0	80.9	83.2	83.5	78.3	68.5	58.7	51.3
	DBStd	14.14	9.98	9.97	8.86	7.27	5.58	3.70	2.88	3.57	5.17	7.62	9.05	9.61
	HDD50	340	127	66	19	1	0	0	0	0	0	1	25	101
	HDD65	1792	478	329	194	61	5	0	0	0	2	54	229	440
	CDD50	6780	124	175	342	519	775	928	1031	1038	850	573	285	140
	CDD65	2759	10	19	52	129	315	478	566	573	402	161	40	14
	CDH74	27679	34	95	343	1016	2811	4970	6361	6487	3852	1418	244	48
	CDH80	12774	0	10	49	237	1075	2386	3272	3455	1803	454	32	1
	WSAvg	6.1	6.7	7.2	7.3	7.4	6.7	5.8	5.1	4.8	4.8	5.4	5.9	6.1
	PrecAvg	44.8	4.1	3.4	3.7	3.5	4.5	4.3	2.7	2.8	4.0	3.7	4.2	4.3
(15) Precipitation	PrecMax	69.7	13.1	10.1	8.3	8.8	12.1	12.9	6.3	8.5	11.0	9.9	16.0	10.1
	PrecMin	26.5	0.2	0.5	0.2	0.3	0.0	0.4	0.0	0.2	0.7	0.1	0.7	0.6
	PrecStd	10.6	2.8	2.1	1.9	2.2	2.9	3.1	1.7	2.2	2.2	2.9	2.9	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.6	81.2	84.3	88.7	93.8	99.6	100.5	103.3	99.1	90.7	83.5	79.0
		MCWB	64.8	65.1	67.7	71.4	74.4	76.1	76.2	74.7	73.9	73.5	70.9	68.5
	2%	DB	74.1	76.9	81.2	85.3	90.8	95.7	97.9	99.3	94.8	88.0	80.0	74.6
		MCWB	63.8	64.8	66.7	70.2	74.1	75.8	76.3	75.9	74.7	72.4	68.6	67.1
	5%	DB	70.5	73.2	78.2	82.7	88.5	93.3	95.3	96.7	92.2	85.3	76.8	71.5
		MCWB	62.7	63.4	65.1	68.7	73.9	76.0	76.8	76.6	74.6	70.5	66.5	65.5
	10%	DB	67.3	70.1	74.7	80.3	86.3	91.0	93.2	94.3	89.7	82.1	73.3	68.2
		MCWB	60.9	62.7	63.8	67.3	73.0	76.2	76.8	76.6	74.1	69.8	66.0	63.2
	0.4%	WB	69.0	69.5	71.2	75.3	78.6	80.4	80.5	80.5	79.4	77.8	73.4	70.9
		MCDB	73.1	75.1	78.9	81.6	88.3	89.5	91.3	91.6	88.6	85.4	79.0	73.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	67.1	68.0	69.5	73.3	77.1	79.2	79.5	79.7	78.3	76.2	71.3	69.1
		MCDB	71.5	73.2	75.7	80.4	86.5	88.7	90.0	90.5	87.3	83.0	76.3	73.1
	5%	WB	65.1	66.5	68.4	71.8	75.9	78.4	78.8	78.9	77.4	74.7	69.7	67.1
		MCDB	69.3	71.3	74.4	78.8	84.4	87.7	89.1	89.5	86.2	80.9	74.3	71.0
	10%	WB	62.5	64.3	66.6	70.5	74.8	77.7	78.2	78.2	76.4	72.9	67.5	64.3
		MCDB	66.3	68.8	72.7	77.0	82.7	86.9	88.3	88.5	84.7	78.5	72.4	67.7
(35) Mean Daily Temperature Range	5% DB	MDBR	21.2	20.9	21.8	22.0	20.1	19.4	20.1	21.1	21.2	22.7	21.9	20.5
		MCDBR	24.4	23.4	23.3	22.4	21.6	22.4	23.3	24.8	24.2	24.2	23.8	22.4
		MCWBR	15.6	13.3	11.4	10.0	8.0	6.6	6.2	6.4	7.8	10.6	13.6	15.3
	5% WB	MCDBR	19.2	18.8	19.0	18.1	18.6	18.7	20.1	20.8	19.4	18.4	19.9	18.7
		MCWBR	14.3	12.9	11.7	10.0	8.3	6.9	6.7	6.7	7.3	9.7	13.6	14.6
(40) Clear-Sky Solar Irradiance	taub		0.326	0.346	0.364	0.410	0.452	0.465	0.482	0.475	0.435	0.370	0.362	0.335
	taud		2.527	2.481	2.427	2.306	2.229	2.240	2.225	2.239	2.312	2.476	2.446	2.512
	Ebn at Noon		287	290	291	280	268	263	258	259	265	278	270	277
	Edh at Noon		27	31	35	42	45	44	45	44	39	31	30	27
(44) All-Sky Solar Radiation	RadAvg		888	988	1378	1693	1856	1969	1980	1873	1602	1328	980	770
	RadStd		81	174	128	108	137	149	148	142	148	181	109	95

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	+162

Nomenclature: See separate page